

Immunomodulatory effects of a standardized *Lycium barbarum* fruit juice in Chinese older healthy human subjects.

Lycium barbarum has been traditionally used in combination with several herbs for medicinal properties, but systematic modern clinical evaluation as a single herb has not been reported. To examine the systematic effects of *L. barbarum* on immune function, general well-being, and safety, we tested the effects of a standardized *L. barbarum* fruit juice (GoChi, FreeLife International, Phoenix, AZ, USA) at 120 mL/day, equivalent to at least 150 g of fresh fruit, the amount traditionally used, or placebo for 30 days in a randomized, double-blind, placebo-controlled clinical study in 60 older healthy adults (55-72 years old). The GoChi group showed a statistically significant increase in the number of lymphocytes and levels of interleukin-2 and immunoglobulin G compared to pre-intervention and the placebo group, whereas the number of CD4, CD8, and natural killer cells or levels of interleukin-4 and immunoglobulin A were not significantly altered. The placebo group showed no significant changes in any immune measures. Whereas the GoChi group showed a significant increase in general feelings of well-being, such as fatigue and sleep, and showed a tendency for increased short-term memory and focus between pre- and post-intervention, the placebo group showed no significant positive changes in these measures. No adverse reactions, abnormal symptoms, or changes in body weight, blood pressure, pulse, visual acuity, urine, stool, or blood biochemistry were seen in either group. In conclusion, daily consumption of GoChi significantly increased several immunological responses and subjective feelings of general well-being without any adverse reactions.